

LID Toolkit Demonstration Report

1. Dataset Overview

The input dataset contains text in **20 languages**, covering a broad range of language families and writing systems:

Arabic (ar), Bulgarian (bg), German (de), Modern Greek (el), English (en), Spanish (es), French (fr), Hindi (hi), Italian (it), Japanese (ja), Dutch (nl), Polish (pl), Portuguese (pt), Russian (ru), Swahili (sw), Thai (th), Turkish (tr), Urdu (ur), Vietnamese (vi), and Chinese (zh).

2. Recommendation

Field	Value
Recommended model	fasttext_subword_exorde-social-media-december-2024-week1
Confidence	67% (2/3 neighbours agree)
Optimised metric	f1_weighted

3. Nearest Neighbour Analysis

The recommendation is derived by comparing the query dataset's linguistic profile against 17 historical datasets using a multi-stratum similarity measure.

Top-3 Most Similar Historical Datasets

Rank	Dataset	Similarity	Best Model	F1 (weighted)	Coverage
1	exorde-social-media-december-2024-week1	21.1%	lid.176_exorde-social-media-december-2024-week1	0.9774	Full
2	language-identification	20.7%	fasttext_subword_exorde-social-media-december-2024-week1	0.9965	Full
3	OpenLID-v2	20.2%	fasttext_subword_exorde-social-media-december-2024-week1	0.9304	Full

Similarity Stratum Distances (Query vs. Nearest Neighbour)

Stratum	Distance
Typological flags (script, tonality, family)	0.3529
Morphological richness	48.6295
Information-theoretic profile	61.8441
Cross-level cohesion	89.3375
Lexical diversity	235.8656
Structural / syntactic structure	351.7579

4. Full Model Rankings

All models ranked by IDW-weighted **f1_weighted** score (coverage-penalised). Models marked **[full coverage]** support all 20 query languages.

Rank	Model	Score	Coverage
1	fasttext_subword_exorde-social-media-december-2024-week1 ← recommended	0.9651	Full
2	tfidf_lr_char_ngram_3_5_exorde-social-media-december-2024-week1	0.9352	Full
3	fasttext_subword_wikipedia	0.9290	Full
4	bow_maxabs_lr_char_ngram_3_5_exorde-social-media-december-2024-week1	0.9250	Full
5	bow_char_ngram_3_5_exorde-social-media-december-2024-week1	0.9238	Full
6	tfidf_char_ngram_3_5_exorde-social-media-december-2024-week1	0.9226	Full
7	fasttext_word_exorde-social-media-december-2024-week1	0.8588	Full
8	fasttext_subword_OpenLID-v2	0.8496	avg gap: 1.0 langs
9	tfidf_lr_char_ngram_3_5_wikipedia	0.8431	Full
10	bow_maxabs_lr_char_ngram_3_5_OpenLID-v2	0.8393	avg gap: 1.0 langs

Rank	Model	Score	Coverage
11	tfidf_lr_char_ngram_3_5_OpenLID-v2	0.8381	avg gap: 1.0 langs
12	bow_char_ngram_3_5_wikipedia	0.8365	Full
13	tfidf_char_ngram_3_5_wikipedia	0.8360	Full
14	bow_char_ngram_3_5_flores_plus	0.8347	avg gap: 1.0 langs
15	xlm_v_base_language_id	0.8306	avg gap: 2.3 langs
16	tfidf_char_ngram_3_5_flores_plus	0.8305	avg gap: 1.0 langs
17	tfidf_char_ngram_3_5_OpenLID-v2	0.8271	avg gap: 1.0 langs
18	bow_char_ngram_3_5_OpenLID-v2	0.8267	avg gap: 1.0 langs
19	fasttext_word_OpenLID-v2	0.8224	avg gap: 1.0 langs
20	bow_maxabs_lr_char_ngram_3_5_wikipedia	0.8193	Full
21	tfidf_lr_char_ngram_3_5_flores_plus	0.8175	avg gap: 1.0 langs
22	fasttext_subword_flores_plus	0.7964	avg gap: 1.0 langs
23	fasttext_word_wikipedia	0.7917	Full
24	bow_maxabs_lr_char_ngram_3_5_flores_plus	0.7744	avg gap: 1.0 langs
25	fasttext_word_flores_plus	0.7183	avg gap: 1.0 langs
26	fasttext_subword_massive	0.6416	avg gap: 4.0 langs
27	tfidf_lr_char_ngram_3_5_massive	0.6195	avg gap: 4.0 langs
28	bow_maxabs_lr_char_ngram_3_5_massive	0.6081	avg gap: 4.0 langs
29	tfidf_char_ngram_3_5_massive	0.6003	avg gap: 4.0 langs
30	bow_char_ngram_3_5_massive	0.5999	avg gap: 4.0 langs
31	fasttext_subword_multilingual_cc_news	0.5761	avg gap: 3.0 langs
32	fasttext_word_massive	0.5473	avg gap: 4.0 langs
33	tfidf_char_ngram_3_5_multilingual_cc_news	0.5303	avg gap: 3.0 langs
34	bow_char_ngram_3_5_multilingual_cc_news	0.5303	avg gap: 3.0 langs
35	tfidf_lr_char_ngram_3_5_multilingual_cc_news	0.5194	avg gap: 3.0 langs
36	fasttext_word_multilingual_cc_news	0.5040	avg gap: 3.0 langs
37	bow_maxabs_lr_char_ngram_3_5_multilingual_cc_news	0.4941	avg gap: 3.0 langs
38	bow_char_ngram_3_5_language-identification	0.4549	avg gap: 6.0 langs
39	tfidf_char_ngram_3_5_language-identification	0.4549	avg gap: 6.0 langs
40	tfidf_lr_char_ngram_3_5_language-identification	0.4544	avg gap: 6.0 langs
41	fasttext_subword_language-identification	0.4535	avg gap: 6.0 langs
42	bow_maxabs_lr_char_ngram_3_5_language-identification	0.4421	avg gap: 6.0 langs
43	bow_char_ngram_3_5_multi_eurlex	0.4378	avg gap: 5.0 langs
44	tfidf_char_ngram_3_5_multi_eurlex	0.4370	avg gap: 5.0 langs
45	fasttext_word_language-identification	0.3994	avg gap: 6.0 langs
46	bow_maxabs_lr_char_ngram_3_5_multi_eurlex	0.3664	avg gap: 5.0 langs
47	fasttext_word_europarl	0.3606	avg gap: 7.0 langs
48	fasttext_subword_multi_eurlex	0.3555	avg gap: 5.0 langs
49	tfidf_lr_char_ngram_3_5_europarl	0.3549	avg gap: 7.0 langs
50	bow_maxabs_lr_char_ngram_3_5_europarl	0.3534	avg gap: 7.0 langs
51	bow_char_ngram_3_5_europarl	0.3428	avg gap: 7.0 langs
52	tfidf_char_ngram_3_5_europarl	0.3428	avg gap: 7.0 langs
53	fasttext_subword_europarl	0.3423	avg gap: 7.0 langs
54	lid.176_exorde-social-media-december-2024-week1	0.3277	Full
55	tfidf_lr_char_ngram_3_5_multi_eurlex	0.3240	avg gap: 5.0 langs

Rank	Model	Score	Coverage
56	fasttext_subword_mmarco	0.3143	avg gap: 8.0 langs
57	bow_maxabs_lr_char_ngram_3_5_stsb_multi_mt	0.3094	avg gap: 8.0 langs
58	tfidf_lr_char_ngram_3_5_stsb_multi_mt	0.3091	avg gap: 8.0 langs
59	tfidf_lr_char_ngram_3_5_mmarco	0.3091	avg gap: 8.0 langs
60	cld3_exorde-social-media-december-2024-week1	0.3067	Full
61	fasttext_subword_stsb_multi_mt	0.3052	avg gap: 8.0 langs
62	bow_maxabs_lr_char_ngram_3_5_mmarco	0.3049	avg gap: 8.0 langs
63	fasttext_word_stsb_multi_mt	0.3049	avg gap: 8.0 langs
64	cld3_OpenLID-v2	0.2939	avg gap: 0.3 langs
65	bow_char_ngram_3_5_mmarco	0.2896	avg gap: 8.0 langs
66	tfidf_char_ngram_3_5_mmarco	0.2895	avg gap: 8.0 langs
67	fasttext_word_mmarco	0.2866	avg gap: 8.0 langs
68	lid.176_OpenLID-v2	0.2822	avg gap: 0.3 langs
69	fasttext_word_multi_eurlex	0.2801	avg gap: 5.0 langs
70	tfidf_char_ngram_3_5_stsb_multi_mt	0.2791	avg gap: 8.0 langs
71	bow_char_ngram_3_5_stsb_multi_mt	0.2789	avg gap: 8.0 langs
72	lid.176_language-identification	0.2213	avg gap: 2.0 langs
73	tfidf_lr_char_ngram_3_5_multilingual_toxicity_dataset	0.2208	avg gap: 9.0 langs
74	cld3_language-identification	0.2188	avg gap: 2.0 langs
75	fasttext_subword_multilingual_toxicity_dataset	0.2155	avg gap: 9.0 langs
76	bow_maxabs_lr_char_ngram_3_5_multilingual_toxicity_dataset	0.2137	avg gap: 9.0 langs
77	tfidf_char_ngram_3_5_multilingual_toxicity_dataset	0.2068	avg gap: 9.0 langs
78	bow_char_ngram_3_5_multilingual_toxicity_dataset	0.2065	avg gap: 9.0 langs
79	fasttext_word_multilingual_toxicity_dataset	0.1797	avg gap: 9.0 langs
80	fasttext_subword_xlsum	0.1572	avg gap: 10.0 langs
81	tfidf_lr_char_ngram_3_5_xlsum	0.1464	avg gap: 10.0 langs
82	tfidf_char_ngram_3_5_xlsum	0.1435	avg gap: 10.0 langs
83	bow_char_ngram_3_5_xlsum	0.1412	avg gap: 10.0 langs
84	bow_maxabs_lr_char_ngram_3_5_xlsum	0.1410	avg gap: 10.0 langs
85	fasttext_word_xlsum	0.1394	avg gap: 10.0 langs
86	bow_maxabs_lr_char_ngram_3_5_xnli	0.1264	avg gap: 11.0 langs
87	fasttext_subword_xnli	0.1259	avg gap: 11.0 langs
88	tfidf_lr_char_ngram_3_5_xnli	0.1252	avg gap: 11.0 langs
89	fasttext_word_xnli	0.1247	avg gap: 11.0 langs
90	tfidf_char_ngram_3_5_xnli	0.1057	avg gap: 11.0 langs
91	bow_char_ngram_3_5_xnli	0.1056	avg gap: 11.0 langs
92	fasttext_word_tweet_sentiment_multilingual	0.0983	avg gap: 12.0 langs
93	bow_maxabs_lr_char_ngram_3_5_tweet_sentiment_multilingual	0.0959	avg gap: 12.0 langs
94	fasttext_subword_tweet_sentiment_multilingual	0.0956	avg gap: 12.0 langs
95	tfidf_lr_char_ngram_3_5_tweet_sentiment_multilingual	0.0938	avg gap: 12.0 langs
96	tfidf_lr_char_ngram_3_5_amazon_reviews_multi	0.0937	avg gap: 12.0 langs
97	fasttext_subword_amazon_reviews_multi	0.0920	avg gap: 12.0 langs
98	bow_char_ngram_3_5_tweet_sentiment_multilingual	0.0916	avg gap: 12.0 langs
99	bow_maxabs_lr_char_ngram_3_5_amazon_reviews_multi	0.0914	avg gap: 12.0 langs
100	tfidf_char_ngram_3_5_tweet_sentiment_multilingual	0.0914	avg gap: 12.0 langs

Rank	Model	Score	Coverage
101	tfidf_char_ngram_3_5_amazon_reviews_multi	0.0883	avg gap: 12.0 langs
102	bow_char_ngram_3_5_amazon_reviews_multi	0.0882	avg gap: 12.0 langs
103	fasttext_word_amazon_reviews_multi	0.0786	avg gap: 12.0 langs
104	bow_char_ngram_3_5_tydiqa	0.0309	avg gap: 14.0 langs
105	tfidf_char_ngram_3_5_tydiqa	0.0302	avg gap: 14.0 langs
106	fasttext_subword_tydiqa	0.0288	avg gap: 14.0 langs
107	tfidf_lr_char_ngram_3_5_tydiqa	0.0269	avg gap: 14.0 langs
108	bow_maxabs_lr_char_ngram_3_5_tydiqa	0.0248	avg gap: 14.0 langs
109	fasttext_word_tydiqa	0.0246	avg gap: 14.0 langs

5. Recommendation Validation Report

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Priority metric: f1_weighted

Datasets evaluated: 17

Aggregate Metrics

Metric	Value
Recommendation accuracy	17.6% (3/17 correct)
Mean Δ f1_weighted	+0.0078
Std Δ f1_weighted	0.0125
Median Δ f1_weighted	+0.0022
Min Δ f1_weighted	+0.0000
Max Δ f1_weighted	+0.0489

Note: Although exact-match accuracy is 17.6%, the mean performance gap is only +0.0078 F1, indicating that the recommended model remains near-optimal even when it does not match the strict ground truth.

Per-Dataset Results

Dataset	Recommended	Ground Truth	Correct	GT Score	Rec Score	Δ f1_weighted
OpenLID-v2	fasttext_subword_OpenLID-v2	fasttext_subword_OpenLID-v2	✓	0.9940	0.9940	+0.0000
amazon_reviews_multi	fasttext_subword_exorde-social-media-december-2024-week1	fasttext_subword_amazon_reviews_multi	X	0.9998	0.9990	+0.0009
europarl	xlm_v_base_language_id	bow_char_ngram_3_5_europarl	X	0.9995	0.9984	+0.0011
exorde-social-media-december-2024-week1	fasttext_subword_exorde-social-media-december-2024-week1	lid.176_exorde-social-media-december-2024-week1	X	0.9779	0.9688	+0.0091
flores_plus	xlm_v_base_language_id	xlm_v_base_language_id	✓	0.9952	0.9952	+0.0000
language-identification	fasttext_subword_exorde-social-media-december-2024-week1	fasttext_subword_language-identification	X	0.9979	0.9958	+0.0021
massive	xlm_v_base_language_id	fasttext_subword_massive	X	0.9974	0.9782	+0.0192
mmarco	xlm_v_base_language_id	fasttext_subword_mmarco	X	0.9975	0.9715	+0.0259
multi_eurlex	fasttext_subword_wikipedia	bow_maxabs_lr_char_ngram_3_5_multi_eurlex	X	0.9998	0.9959	+0.0039
multilingual_cc_news	fasttext_subword_exorde-social-media-december-2024-week1	fasttext_subword_multilingual_cc_news	X	0.9971	0.9860	+0.0112
multilingual_toxicity_dataset	fasttext_subword_exorde-social-media-december-2024-week1	fasttext_subword_multilingual_toxicity_dataset	X	0.9937	0.9899	+0.0037

Dataset	Recommended	Ground Truth	Correct	GT Score	Rec Score	Δ f1_weighted
stsb_multi_mt	xlm_v_base_language_id	tfidf_lr_char_ngram_3_5_stsb_multi_mt	X	0.9942	0.9912	+0.0029
tweet_sentiment_multilingual	xlm_v_base_language_id	bow_char_ngram_3_5_tweet_sentiment_multilingual	X	0.9981	0.9492	+0.0489
tydiqa	xlm_v_base_language_id	fasttext_subword_tydiqa	X	0.9999	0.9986	+0.0013
wikipedia	fasttext_subword_wikipedia	fasttext_subword_wikipedia	✓	0.9964	0.9964	+0.0000
xlsum	fasttext_subword_exorde-social-media-december-2024-week1	fasttext_subword_xlsum	X	0.9989	0.9986	+0.0002
xnli	fasttext_subword_exorde-social-media-december-2024-week1	fasttext_subword_xnli	X	0.9996	0.9975	+0.0022